

2. Working with Claude

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Faculty Workshop

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Today's Session

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Academic Studio



2

The Command Line



3

**CLAUDE.md and
Skills**



4

**Claude as a
Teaching Assistant**

Academic Studio

Where Academic Studio Fits

- Original Claude Code was CLI (command line interface)
- Millions of developers use Claude Code extension for VS Code
- VS Code = integrated development environment (IDE) from Microsoft

Academic Studio = VS Code, except

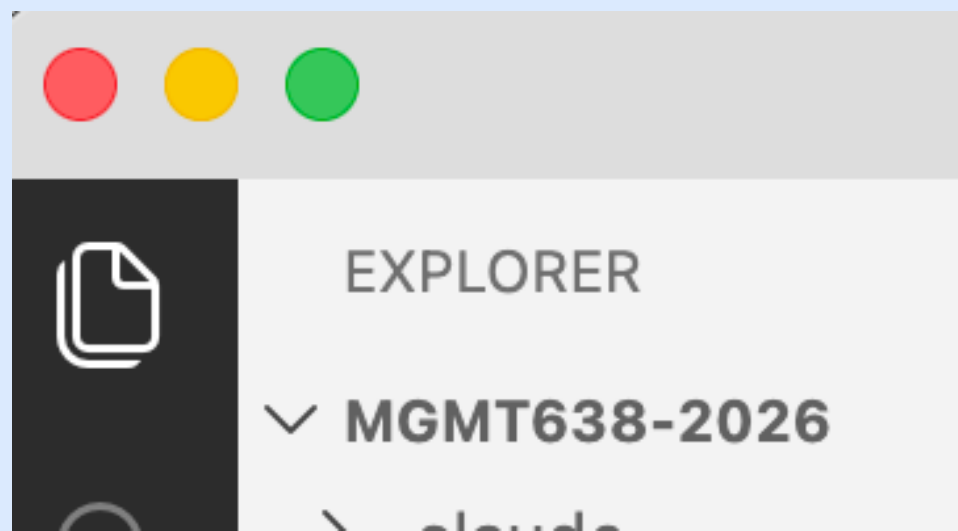
- Removed menus and toolbar items aimed at developers
- Claude Code opens on startup
- Other extensions pre-installed
- New Claude menu and revised Help menu

File Explorer and Search

Reach the File Explorer tool from the left toolbar.

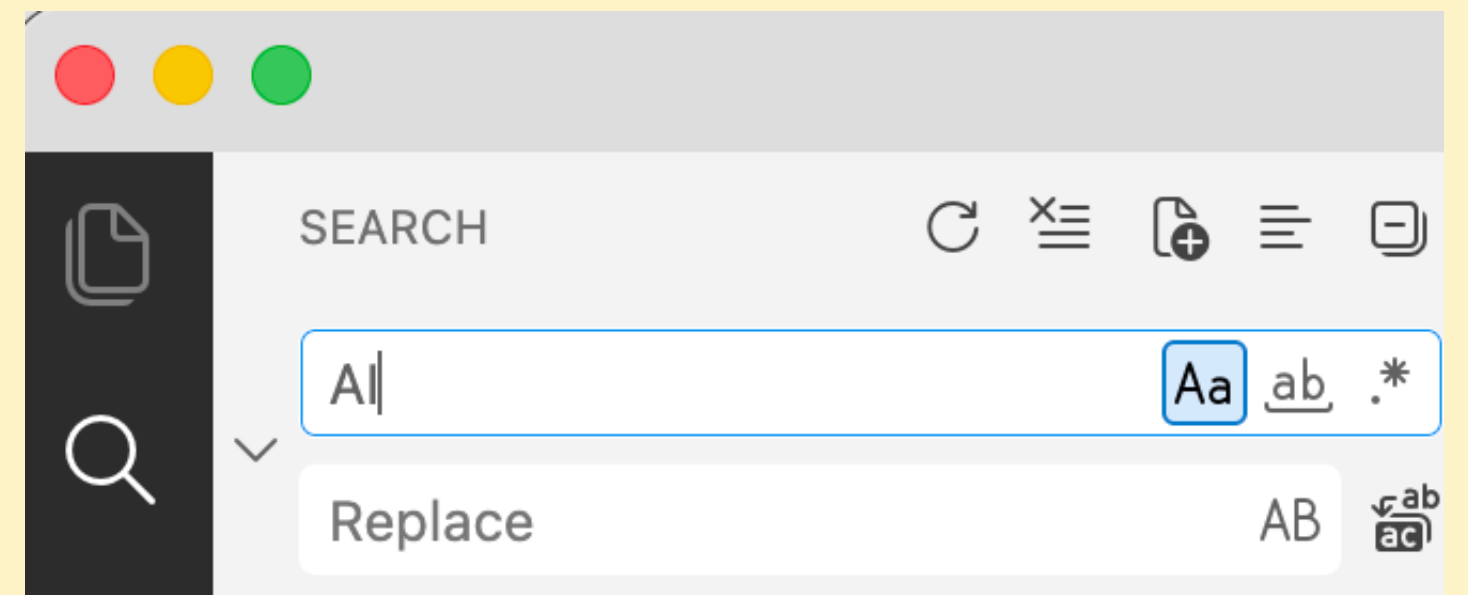
Files

- Right-click to rename, copy, ...
- Double-click to open in Academic Studio viewer
- Viewer provides different options for different file types (cannot edit Office docs)



Search

- The magnifying glass searches **inside every file** in the folder at once
- Options for case, whole word, and regular expressions
- Replace across all files from the same pane



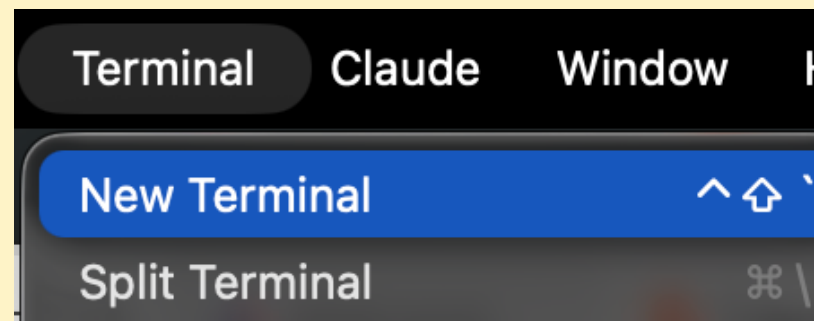
Getting Around

Conversations

- Open a second Claude conversation for a separate task
- Each keeps its own context; they do not see each other

Terminal

- Open from the Terminal menu, X to close
- Claude uses it too — you will see commands run as it works



Remote SSH

- Connect to jgsrc1 or a different remote server
- Files and code execution on remote server. Academic Studio runs on your computer.

Extensions

VS Codium marketplace (all are free)

Permissions

Claude Permissions

How much can Claude do without asking you first? Your choice takes effect when a new chat is

Ask before changes

Claude asks your permission before editing files or running commands.

Allow file edits

Claude edits files without asking, but still asks before running commands.

Plan first

Claude only reads and proposes a plan; nothing changes until you approve it.

Allow everything current setting

Claude edits files and runs commands without ever asking. Use with care.

Some Claude Commands

Command	What it does
<code>/resume</code>	a version of Claude/Conversations
<code>/model</code>	select from Fable, Opus (default), Sonnet, and Haiku
<code>/clear</code>	clear the context window (= new conversation)
<code>/compact</code>	summarize the conversation in a text file, launch a new conversation that reads the text file
<code>Esc</code>	escape key interrupts Claude, which then waits for new instructions

Fit with VS Code

- Independent apps, individual extensions and settings
- They share Claude — same CLAUDE.md, same skills, same conversation history
- Example: Work on project in Academic Studio. Then open in VS Code. [/resume](#) brings up conversation history from Academic Studio.

The Command Line

Using a Terminal (Command Line)

- Windows: Command Prompt, Windows PowerShell, Windows Terminal
- Mac Terminal
- Open. Then type commands. Syntax must be right (not intelligent)

What Claude Can Do at the Command Line

File operations

Move, copy, rename, delete, search within.

Find all files in my Documents folder that discuss ... and move them to ...

Install and run software

Run a python script, build a LaTeX doc, ...

Interact with external services

Command Line Interfaces (CLIs): gh for GitHub, aws for Amazon Web Services, ... Also, use APIs.

More Things Claude Can Do at the Command Line

- `pandoc` — convert between Word, Markdown, HTML, PDF
- `ImageMagick` — resize, crop, convert images in bulk
- `ffmpeg` — trim video, transcode, pull the audio out
- `ripgrep` — search thousands of files in seconds
- `cron` — run a command on a schedule

Example: Git (Search the Past)

Git is a version control system. A **commit** records the current state of a folder (repository=repo). Git **is not** AI. Git existed long before AI. But AI knows Git.

Tell Claude

```
Install Git. Commit this folder to Git.
```

Frequently, thereafter, tell Claude to commit changes.

At some point, you realize that you want to recover something from a version that was overwritten sometime in the past:

Tell Claude

Find a prior version of ... that said ... Add that to the current version.

CLAUDE.md and Skills

CLAUDE.md

Where Claude Looks

- At the start of each conversation, Claude looks in specific places for CLAUDE.md files
- It looks in your home directory (C:\users\yourname on Windows) and it looks in the project folder

What It Is

- A CLAUDE.md is a text file that contains instructions for Claude
- Claude reads the entire file, so use it sparingly

You could get the same behavior by pasting the instructions in the prompt window

Skills

At Startup

At the start of each conversation, Claude also looks in specific places for SKILL.md files

Two Locations

It looks in your home directory (C:\users\yourname on Windows) and it looks in the project folder

Loaded on Demand

Skills are text files saved on your hard drive. Claude reads the top of each SKILL.md file (the skill description) on startup, and it reads the bulk of the file **only when needed**

Installing Skills

Where to Find Them

- There are thousands of skills published on the internet.
- Anthropic publishes skills (xlsx, etc.) at <https://github.com/anthropics/skills/>

Ask Claude

- To search for skills, ask Claude. To install a skill you want, ask Claude.
- Other than a few of Anthropic's skills, the most useful skills are generally those you create, because they are customized to your work.

Example: xlsx Skill

The xlsx, docx, and pptx skills are created by Anthropic and come pre-installed with Claude Desktop

```
---
name: xlsx
description: "Use this skill any time a spreadsheet file is the primary input or output. This means any task where the user wants to: open, read, edit, or fix an existing .xlsx, .xslm, .csv, or .tsv file (e.g., adding columns, computing formulas, formatting, charting, cleaning messy data); create a new spreadsheet from scratch or from other data sources; or convert between tabular file formats. Trigger especially when the user references a spreadsheet file by name or path – even casually (like \"the xlsx in my downloads\") – and wants something done to it or produced from it. Also trigger for cleaning or restructuring messy tabular data files (malformed rows, misplaced headers, junk data) into proper spreadsheets. The deliverable must be a spreadsheet file. Do NOT trigger when the primary deliverable is a Word document, HTML report, standalone Python script, database pipeline, or Google Sheets API integration, even if tabular data is involved."
license: Proprietary. LICENSE.txt has complete terms
---

# Requirements for Outputs

## All Excel files

### Professional Font
- Use a consistent, professional font (e.g., Arial, Times New Roman) for all deliverables unless otherwise instructed by the user

### Zero Formula Errors
- Every Excel model MUST be delivered with ZERO formula errors (#REF!, #DIV/0!, #VALUE!, #N/A, #NAME?)

### Preserve Existing Templates (when updating templates)
- Study and EXACTLY match existing format, style, and conventions when modifying files
- Never impose standardized formatting on files with established patterns
- Existing template conventions ALWAYS override these guidelines
```

Some Examples

I've looked at several skill repos for business/economics research. Most are not useful. Here are a couple I've adopted — by Lu Han, Bank of Canada.

- **Econ Writing**: tells Claude to follow Cochrane's Writing Tips for PhD Students and some other style guides
- **Econ Paper Review**: writes a referee report on your paper

Give Claude the links and ask Claude to install them if you want to try them out.

Creating Skills

1

Iterate

Usual process is you work on something, iterating until you get what you want.



2

Recognize

You realize you may want the same thing again, and you don't want to go through the iteration again.



3

Save the Recipe

Solution: tell Claude to save the satisfactory method (recipe) as a skill. Can save your result along with the skill as an example to follow if useful.

Invoking Skills

Slash Command

- Type `/skillname` at the prompt
- Example: `/plan`
- Invokes the skill directly

Natural Language

- Describe what you need in plain English
- Example: "Use your planning mode and evaluate ..."

Automatic

- Claude may invoke a skill on its own
- Example: "Create an Excel file" triggers the `xlsx` skill automatically
- Driven by the skill's description

Example: Create and Invoke

1

Create

Tell Claude to create a skill
for your folder named
"pirate" with this content:
"Always answer like a pirate"



2

Invoke

Start a new conversation.
Enter `/pirate` **What is the
meaning of life?**



3

Inspect

Ask Claude to show you the
pirate SKILL.md

Claude as a Teaching Assistant

What can Claude Do?

- Draft slides
- Assist with editing slides
- Generate images for slides
- Draft quizzes and assignments
- Assist with grading

- Draft emails to students
- Help create async materials
- Build tutors
- Build Excel workbooks
- Write cases

Drafting and Editing Slides

Claude is generally able to create slide decks using whatever style you like. Here is an example.

- `SamplePresentation.pptx` is a Microsoft template
- I asked Claude to remake `tesla.pptx` in that style
- The first draft: `tesla1.pptx`
- The second draft: `tesla2.pptx`

Creating a Skill

1

Save the Recipe

Once Claude has been successful at creating decks in your preferred style, ask Claude to save the recipe as a skill.



2

What Gets Saved

Claude will save images, layout guides, etc. from the doc you gave it along with the fonts that best match the doc (maybe not exact match) for reuse.

Isn't This Just a PowerPoint Template?

Partly, yes — and a good skill does not replace the template.

What the Template Already Does

Fonts, colors, layouts, placeholder positions, the scenery. A designed .pptx is already a style kit.

A good skill **uses** it — filling that template's own layouts instead of drawing shapes.

What the Skill Adds

- Which layout for which kind of content
- How much text fits in each placeholder
- What to do when nothing fits — a wide chart, a long table

A template guides **a person**, who sees the result and adjusts. A skill guides **Claude**, which cannot.

Claude in PowerPoint

In May, 2026, the Claude add-in for PowerPoint became generally available (for paid accounts). But I don't think Rice has authorized it for our Microsoft 365 accounts.

The advantages would be (1) just one window open, (2) no save/close/reopen.

Tips for Your Skill

Save and close first

Include in your skill:
Save and close any open PowerPoint deck before editing it. Re-open after editing.

Claude cannot see your slides

Claude cannot natively view PowerPoint slides to examine whether things are properly located. It goes PPTX → PDF → PNG.

The route matters

If left to its own devices, it may choose an inferior PPTX → PDF route (LibreOffice = open-source version of MS Office)

When Claude creates your skill, ask it about using AppleScript (Mac) or COM automation (Windows)

One More Tip

Sometimes multiple skills compete for the same job.

The collision

Anthropic's pptx skill triggers on **any** PowerPoint file, so a skill of your own for your desired style will collide with it. Things compound if you create multiple skills for multiple styles.

The fix

Ask Claude to edit your CLAUDE.md to settle the precedence.

Exercises

1. Tell Claude to install Git and to initialize your project folder as a Git repo and to commit. Make some changes, commit again, and ask about restoring the prior version.
2. Pick a deck you want to refresh. Ask Claude to read it and offer suggestions on content, organization, and style. Ask Claude to edit some slides and to draft some slides, saving changes in a new name.
3. Create a skill for Claude to write slides in your desired style. Inspect the skill (if global, it is in `$HOME/.claude/skills`; if local, it is in your project folder `.claude/skills`)

Takeaways

Claude can work alongside you on whatever you're working on. Get advice and suggestions. Hand over the tedious part of the work. Save good workflows as skills so they can be easily repeated.